Natural Language Processing: Assignment 2

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1 Dataset & Setup

To classify news articles into one of four categories (world, sports, business, and sci/tech), we use the AG News dataset from Hugging Face [Zhang et al., 2015]. This dataset consists of news articles with associated labels, titles, and descriptions. Each class contains 30,000 training samples and 1,900 testing samples, resulting in a total of 120,000 training samples and 7,600 testing samples.

We used the official Hugging Face split (120,000 train and 7,600 test). The training set was split into a new training set (of size 108,000) and a development set (of size 12,000) using stratified sampling with a fixed random seed of 67 to ensure reproducibility. The final split sizes are: train 108,000, dev 12,000, and test 7,600.

Preprocessing. We combined the title and description of each news article into a single text input. We cleaned the text by HTML unescaping to convert HTML entities back to their original characters (like `#39;` to `'`) and removed any remaining HTML tags. We also unescaped LaTeXentities (\\ etc.), collapse backslashes and normalize whitespace. We converted the labels from 1-based to 0-based indexing (i.e., world: 0, sports: 1, business: 2, sci/tech: 3) for easier handling in the classification models.

Tokenization. We used Byte Pair Encoding (BPE) tokenization, which is a subword tokenization method that can effectively handle out-of-vocabulary words and reduce the vocabulary size while still capturing meaningful subword units. We trained a BPE tokenizer training set with a vocabulary size of 5,000 and a minimum frequency of 2 to ensure that only subwords that appear at least twice in the training data are included in the vocabulary. We convert token sequences to fixed-length inputs via truncation/padding to a maximum sequence length (64 vs. 128 vs. 256).

2 Models

In this assignment, we compare two neural models for news classification: a Convolutional Neural Network (CNN) and a Long Short-Term Memory network (LSTM). To make the comparison fair, both models use the same data splits, preprocessing pipeline, and evaluation setup.

Shared input representation. Both models receive the same preprocessed and tokenized text (see Section 1 for preprocessing and tokenization details). Sequences are padded or truncated to the chosen maximum length. Both models begin with a trainable embedding layer that maps token ids to dense vectors. Note that the CNN has inherent feature extraction through its convolutional layers, while the LSTM relies on the embeddings and recurrent connections for representation learning.

CNN model. The CNN is based on the text classification model of [Kim, 2014]. After the embedding layer, the model applies several 1-dimensional convolution filters in parallel. These filters are used to detect informative local patterns in the text, such as short phrases or combinations of nearby words. We use multiple kernel sizes so the model can capture patterns of different lengths. After the convolution step, a ReLU activation function is applied, followed by max-pooling to keep the strongest signal from each filter. The resulting features are then concatenated, followed by dropout for regularisation, and finally passed to a linear layer that predicts one of the four classes.

LSTM model. The LSTM model is based on the Long Short-Term Memory architecture introduced by [Hochreiter and Schmidhuber, 1997]. It reads the input sequence token by token, which makes it suitable for modelling word order and longer dependencies in the text. We use a stacked 2-layer LSTM. To avoid learning from padding tokens, we handle variable-length sequences by skipping over padded positions during the recurrent computation, so the model only processes the real tokens in each example. For classification, we use the final hidden state of the top LSTM layer as the representation of the whole article. In the bidirectional setting, the model combines information from both directions before making the final prediction. As in the CNN, dropout is applied before the final linear layer.

Hyperparameters. Table 1 shows the hyperparameter search space for both models. For each model, the best setting was selected based on macro-F1 on the development set.

Parameter	**CNN**	**LSTM**
Embedding dim d	{128, 512}	{128, 256}
Filters F	{100, 300}	—
Kernel sizes K	{[3,5], [3,5,7]}	—
Hidden dim H	—	{128, 256, 512}
Bidirectional	—	{False, True}
Dropout	{0.3, 0.5}	0.5
Learning rate	{1e-3, 1e-4}	{1e-3, 1e-4}
Weight decay	{1e-5, 1e-3}	{0, 1e-5}

Table 1: Hyperparameter search spaces for the CNN and LSTM models.

Shared training settings. Both models were trained with the Adam optimiser. We used gradient clipping with a maximum norm of 1.0 and early stopping with a patience of 3 epochs based on validation loss. The batch size was set to 512, sequence length to 128, and the final training runs lasted for at most 15 epochs. During hyperparameter tuning, each configuration was trained for 5 epochs.

Ablation: maximum sequence length. As our ablation study, we vary the maximum sequence length and compare settings of 64, 128, and 256 tokens. To clarify, the hyperparameters in Table 1 were only tuned for the seq128 setting. For the other two sequence lengths, we kept all hyperparameters fixed to the best values found in the seq128 runs. This allows us to study how much context the models need and whether giving them longer input sequences improves performance enough to justify the extra computation.

3 Experimental Protocol

For each model, we first performed hyperparameter tuning on the training set. The CNN and LSTM were tuned separately using grid search over the parameter ranges shown in Table 1 with a fixed maximum sequence length of 128 tokens. During tuning, each configuration was trained for 5 epochs, and the best

configuration was selected based on macro-F1 on the development set. Early stopping with a patience of 3 epochs was enabled during this tuning stage, although in practice the short tuning runs meant that most configurations finished all 5 epochs without triggering early stopping.

After selecting the best hyperparameters for a given model, we trained that model again on the train data and evaluated it on the test set. The final training runs used the Adam optimiser, a batch size of 512, gradient clipping with a maximum norm of 1.0, and a maximum of 15 epochs. Unlike the tuning stage, the final training procedure did not apply early stopping; instead, models were trained for the full configured number of epochs and then evaluated on the test set. We did this to ensure that the final models had the opportunity to fully converge and to allow us to generate learning curves for later analysis.

We report accuracy and macro-F1 as our main evaluation metrics, with macro-F1 used as the primary measure for model selection because it gives equal weight to all four classes. In addition, we generated classification reports, confusion matrices, learning curves, and files with misclassified test examples for later error analysis. The test set was only used for final evaluation after model selection.

As our ablation study, we varied the maximum input sequence length while keeping the rest of the setup fixed. We ran each model with maximum sequence lengths of 64, 128, and 256 tokens, resulting in six main experiment settings: CNN-64, CNN-128, CNN-256, LSTM-64, LSTM-128, and LSTM-256. This design allows us to compare the two architectures directly and also study how much performance depends on the amount of input context available to the model.

4 Results

4.1 Best hyperparameters

Table 2 summarises the best hyperparameters from the dedicated tuning runs only (`experiment_cnn_tuning` and `experiment_lstm_tuning`). The CNN preferred a large embedding dimension (512), a high number of filters (300), and a learning rate of 0.001. The LSTM preferred a larger hidden dimension (512), embedding dimension 256, bidirectional encoding, and a learning rate of 0.001. Both architectures benefited from very low weight decay (1e-5 or 0.0).

Model	**Embed**	**Filters**	**Hidden**	**BiLSTM**	**LR**	**Dropout**	**WD**
CNN	512	300	—	—	0.001	0.3	1e-5
LSTM	256	—	512	Yes	0.001	—	0.0

Table 2: Best hyperparameter configurations from dedicated tuning runs (`experiment_cnn_tuning` and `experiment_lstm_tuning`). Embed: embedding dimension, WD: weight decay.

4.2 Test set performance

Figure 1 shows the distribution of dev macro-F1 scores for both models across all hyperparameter search trials. The distributions are relatively tight, especially for the LSTM, indicating that the tuning process was stable and not overly sensitive to hyperparameter choices. The CNN shows a slightly wider distribution, suggesting that its performance may be more sensitive to hyperparameter settings than the LSTM. Overall, both models achieved high dev macro-F1 scores across a range of hyperparameters, which is a positive sign for their generalization capabilities.

The test results are shown in Table 3. At the main comparison sequence length of 128 tokens, the CNN slightly outperformed the LSTM, achieving a test accuracy of 90.71% and a macro-F1 of 90.69%, compared

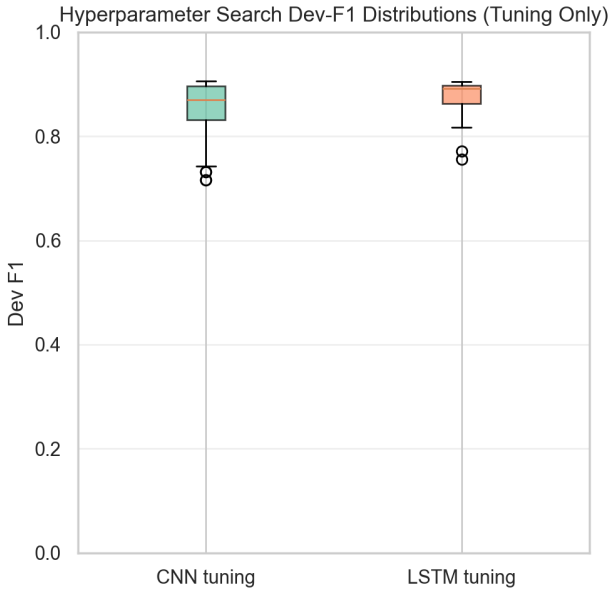

Figure 1: Distribution of dev macro-F1 scores across all hyperparameter search trials, grouped by model. Higher and tighter distributions indicate more stable tuning.

to the LSTM's accuracy of 90.57% and macro-F1 of 90.59%.

Interestingly, the ablation results reveal that longer sequences do not consistently improve performance. For the CNN, the best test macro-F1 was actually achieved at sequence length 64 (90.96%), while the highest dev F1 was at 256 (91.09%). The LSTM showed a more monotonic trend, improving from 90.32% at seq64 to 90.74% at seq256. The gap between the two models was relatively small across all sequence lengths, with a difference of at most 0.64 percentage points in macro-F1 (at seq64).

Model	**Seq Len**	**Dev F1**	**Test Acc**	**Macro F1**
Main model comparison (seq length = 128)				
CNN	**128**	0.9101	0.9071	0.9069
LSTM	**128**	0.9090	0.9057	0.9059
Ablation: sequence length				
CNN	64	0.9096	**0.9099**	**0.9096**
CNN	256	**0.9109**	0.9075	0.9072
LSTM	64	0.9086	0.9034	0.9032
LSTM	256	0.9076	0.9074	0.9074

Table 3: Main comparison at sequence length 128 (CNN vs. LSTM), followed by sequence-length ablations. Dev F1 is the best development macro-F1 found during tuning. Best scores per metric column are bolded.

4.3 Ablation study: effect of maximum sequence length

Figure 2 shows how test accuracy and macro-F1 vary with sequence length.

The overall trend is clear: The CNN does not benefit from longer sequences, with the best test macro-F1 at seq64. The LSTM, on the other hand, shows a consistent improvement with longer sequences, achieving its best test macro-F1 at seq256. This suggests that the LSTM is better able to leverage the additional context

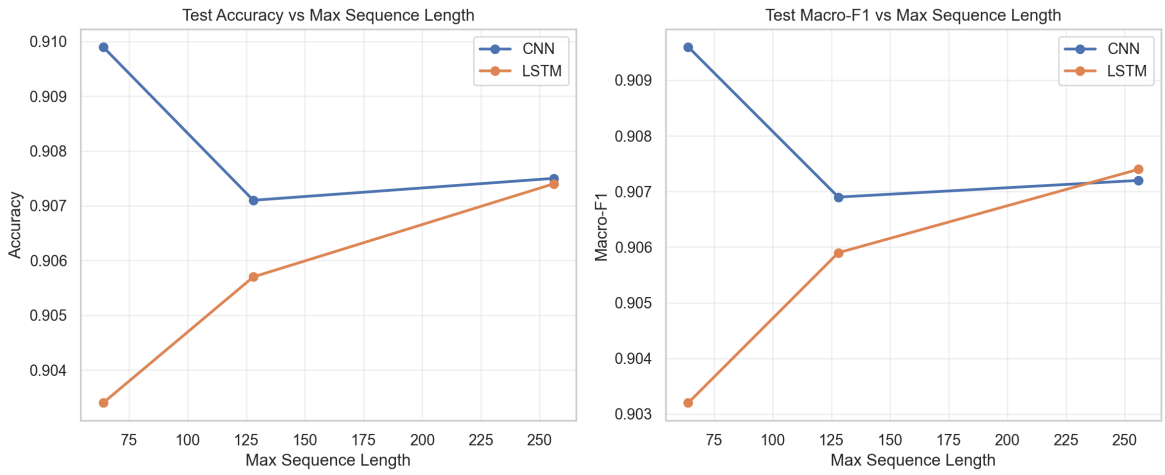

Figure 2: Test accuracy and macro-F1 as a function of maximum sequence length for CNN and LSTM.

provided by longer input sequences, while the CNN may be more prone to overfitting or may not effectively utilize the extra information. This is especially evident by the fact that the CNN outperforms the LSTM at seq64, but is outperformed by the LSTM at seq256.

4.4 Class-wise analysis

Figure 3 presents the per-class macro-F1 scores. The confusion matricies per model and sequence length are shown in Appendix B along with the full classification reports.

The World and Sports classes are generally easier to classify, with F1 scores above 90% in most configurations. The Sci/Tech class is slightly harder, with F1 scores typically around 87–88%. The most challenging class is Business, where F1 scores range from 86% to 87%, depending on the model and sequence length.

4.5 Learning curves

Figure 4 displays the learning curves for the final training runs on the training set and validation set for all six runs (CNN and LSTM at seq64, seq128, and seq256) with the tuned hyperparameters from the dedicated tuning runs. Both training and validation curves are shown for loss and accuracy. The F1 curves are shown in Appendix C.

Both models show rapid convergence, with validation loss decreasing sharply in the first few epochs and then stabilizing. Both models converge to similar final validation accuracies at a similar number of epochs (e.g., around 1–2 epochs). The CNN shows a slightly more stable validation loss curve, while the LSTM shows signs of overfitting after epoch 4, where the validation loss starts to increase while training loss continues to decrease. Most notably, the LSTM at seq256 converges more rapidly as indicated by the steepest initial drop in validation loss, which may explain its superior performance at this sequence length. However, all models converge to similar final validation accuracies.

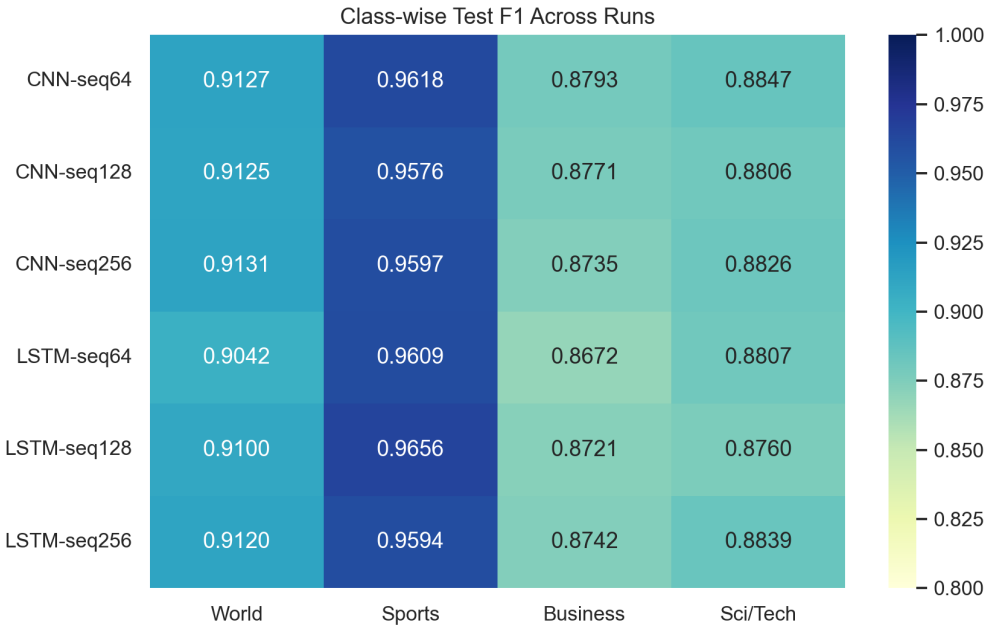

Figure 3: Heatmap of per-class test F1 scores across all runs. World and Sports are easier to classify; Business is the most challenging.

4.6 Summary

The LSTM model achieves the best overall result at the longest sequence length (seq256), with a test macro-F1 of 90.74%, outperforming the CNN's best macro-F1 of 90.72% at seq64. The LSTM benefits more from longer sequences, while the CNN performs best at shorter sequences. Both models show stable tuning and learning trajectories, with the Business class being the most challenging to classify.

5 Error Analysis

To better understand the remaining errors, we examined the first 20 misclassified test examples saved for each of the six runs. Rather than looking at isolated mistakes, we focused on examples that reappeared across multiple models and sequence lengths. This makes it possible to identify systematic weaknesses instead of noise from a single run. Full listings of all misclassified examples are provided in Appendix A.

The clearest pattern is confusion between *Sci/Tech* and *Business*. Across the saved error files, *Sci/Tech* articles misclassified as *Business* is the single most common error type, followed by the reverse. This is unsurprising: many technology articles in AG News employ business language such as *market*, *shares*, *growth*, or *workers*, and frequently reference company names. Sci/Tech articles that were misclassified as Business include "IBM to hire even more new workers" (App. A.1, Example 5), "Dell Exits Low-End China Consumer PC Market" (App. A.1, Example 8), and a Google IPO article (App. A.1, Example 6). Conversely, Business articles mislabeled as Sci/Tech include "Intel to delay product aimed for high-definition TVs" (App. A.1, Example 11), "Yahoo! Ups Ante for Small Businesses" (App. A.1, Example 15), and "Surviving Biotech's Downturns" (App. A.1, Example 13). These systematic errors reveal that the boundary between technological innovation and economic reporting is inherently ambiguous, especially in news data.

Another recurring pattern is confusion between *World* and *Business*. Articles about politics or international

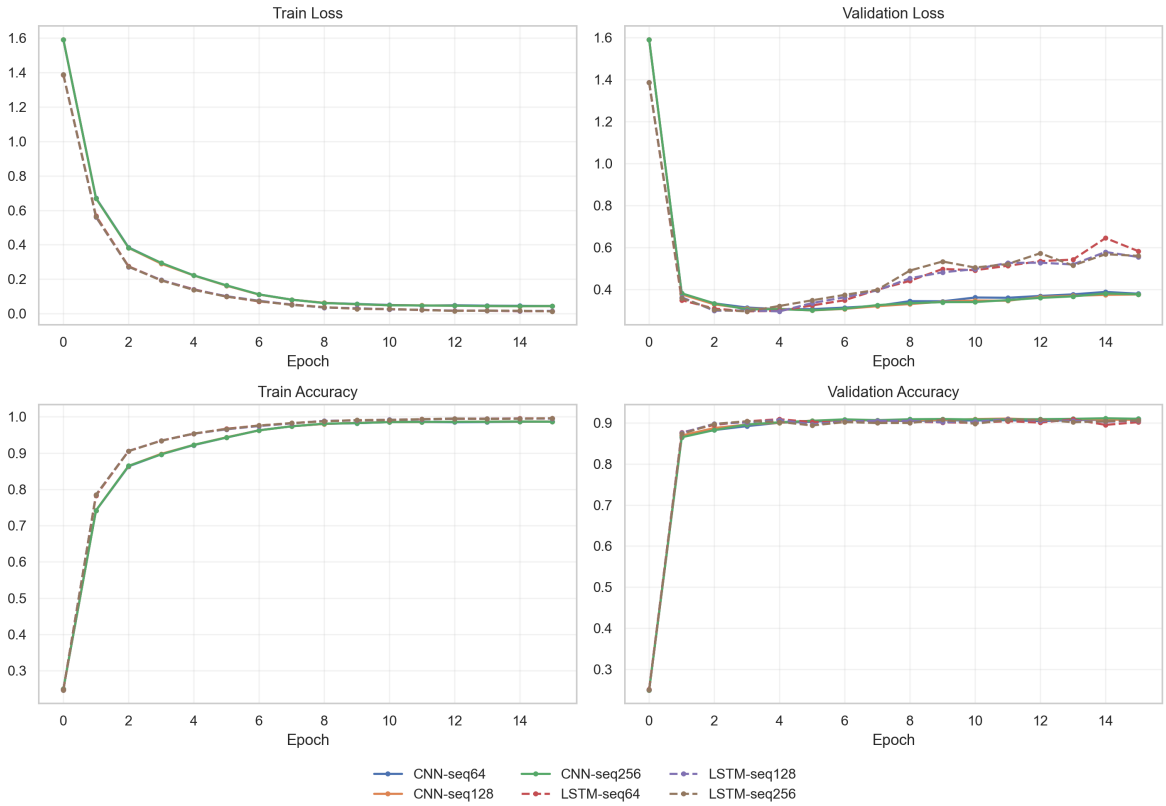

Figure 4: Training and validation loss/accuracy curves for all six runs. Both models show rapid early convergence and stable learning trajectories.

affairs are often misclassified as *Business* when they mention oil prices, trade, or financial consequences. Examples include "Oil prices bubble to record high" (App. A.1, Example 16), "Venezuela Prepares for Chavez Recall Vote" (App. A.2, Example 7), and "Google Lowers Its IPO Price Range" (App. A.1, Example 18). In these cases, the model seems to rely strongly on economic keywords, even when the broader topic is political or international. This suggests that topical overlap between global affairs and economics remains difficult for both architectures.

We also observe a smaller but consistent confusion between *World* and *Sports*. For instance, "Live: Olympics day four" (App. A.1, Example 10) is repeatedly predicted as *Sports* although it is labelled as *World*, and political or public-affairs articles such as "Mr Downer shoots his mouth off" (App. A.1, Example 14) are sometimes predicted as *Sports*. These mistakes suggest that short headlines without much context can be difficult to classify, especially when they contain named entities or event-style phrasing that does not clearly reveal the topic.

Some errors are caused by lexical ambiguity inside very short texts. The article "E-mail scam targets police chief" (App. A.2, Example 2) was misclassified as *World* in multiple runs, probably because words such as *police* and *chief* are also common in political or crime-related world news. Similarly, "Super ant colony hits Australia" (App. A.1, Example 3) was often predicted as *World*, even though it belongs to *Sci/Tech*. In these examples, the model appears to focus on surface cues such as country names or institutional words rather than the underlying scientific content.

Comparing the two models, the overall error types are very similar, but the CNN tends to make slightly fewer mistakes on the hardest category boundaries. The LSTM shares most of the same repeated errors, especially the *Sci/Tech–Business* confusions, which suggests that additional sequence modelling alone is not enough

to solve these cases. Many of the problematic examples are difficult because they are genuinely mixed-topic articles rather than simple classification failures.

The saved errors also show that several examples are misclassified in almost every run. Articles about IBM, Yahoo!, Intel, oil prices, and Google IPOs appear repeatedly across both models and all sequence lengths, indicating that these cases reflect persistent ambiguities in the dataset rather than random noise from individual runs.

Compared with Assignment 1 (TF-IDF with Linear SVM and Logistic Regression), the overall error profile is very similar. In both assignments, the dominant confusion is still *Sci/Tech* vs. *Business*, with *World* vs. *Business* as another recurring boundary. This consistency suggests that the main difficulty is not specific to one classifier family, but to the dataset itself: many headlines combine technological and financial language, or mix political and economic framing. In other words, moving from sparse TF-IDF features to neural sequence models improves scores, but it does not remove the same core ambiguities.

Overall, the error analysis suggests that the main challenge in this task is separating articles that sit near the boundary between two related topics. In particular, the distinction between *Sci/Tech* and *Business* remains the most difficult for both models. This is consistent with the class-wise results reported in Section 4, where Business and Sci/Tech were the lowest-performing categories.

6 Discussion

Our results suggest that this task depends a lot on short local patterns in the text. That helps explain why the CNN is already strong at sequence length 64: even with shorter inputs, it can pick up useful topic phrases and keywords. So the CNN advantage at short length is not only a score difference, but also a good match between the model and the kind of signal in AG News.

At the same time, the BiLSTM improves more as we increase sequence length, and at 256 tokens it almost catches the CNN (Table 3). A likely reason is that the extra context helps the LSTM in examples where early words are not enough to decide the class. This seems especially relevant for Business and Sci/Tech, where articles often contain mixed vocabulary. Still, even at 256 the LSTM does not clearly pass the CNN, which suggests that local cues remain very important for this dataset.

One result that we should explicitly discuss is the CNN-128 anomaly. In Table 3, CNN-128 is slightly worse than CNN-64 on test accuracy and macro-F1 (0.9049 vs. 0.9061 and 0.9047 vs. 0.9059). So the sequence-length ablation is not perfectly monotonic. We cannot fully prove the cause from one run, but the difference is very small (about 0.1 points) and could come from normal tuning/run variance together with a less favorable hyperparameter setting for that run.

The class-wise results support this broader picture. World and Sports are generally easier for both models, while Business and Sci/Tech remain harder. This tells us that the remaining errors are mostly about separating semantically close topics, not about basic model failure.

Overall, CNN-seq256 is still the best single configuration in our experiments, but the margin over LSTM-256 is small. However, although the CNN is better, it does not scale with sequence length as well as the LSTM, we think that if sequence length were increased further, the LSTM might eventually outperform the CNN. This is because the LSTM can potentially leverage longer-range dependencies and more global context, while the CNN relies more on local patterns that may saturate in usefulness after a certain point. In future work, it would be interesting to explore even longer sequence lengths and see if the LSTM can eventually surpass the CNN as it gets more context to work with.

7 Limitations

This study has a few limitations. First, we only evaluate on AG News, so our conclusions may not transfer to datasets with different writing styles or label structures. A second limitation is our choice of a 5,000-item BPE vocabulary. This is a reasonable size for a small study, but we did not check token coverage on the test set, so we cannot say how much information may have been lost because of vocabulary size. A larger vocabulary might have preserved more useful distinctions, especially for named entities and domain-specific terms.

Another important limitation is the tuning setup. Hyperparameter search was done with only 3 epochs per configuration, and we did not explicitly verify that this was long enough for strong and weak configurations to separate clearly. This means some settings may have looked worse simply because they learned more slowly, not because they were truly worse. The CNN-128 result also shows that small differences between runs should be interpreted carefully.

Regularisation is another place where our choices may have been too limited. In the best configurations, weight decay was either very small or zero, and the learning curves suggest that the LSTMs overfit slightly more than the CNNs. Because of this, it would have been useful to test higher weight decay values more aggressively, especially for the LSTM models.

Finally, we compare only CNN and LSTM models trained from scratch, so this is a comparison between two classical neural architectures rather than a comparison with the current strongest text-classification models. That does not affect the internal comparison in this report, but it does limit how far the absolute performance numbers can be taken.

References

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[Zhang et al., 2015] Zhang, X., Zhao, J., and LeCun, Y. (2015). Character-level convolutional networks for text classification. In *Advances in Neural Information Processing Systems 28 (NIPS 2015)*.

A Detailed Misclassification Examples

This appendix provides a listing of misclassified examples from the test set across all six experimental runs. Each example is labeled with its dataset index, gold label, and predicted label for reference during error analysis.

A.1 CNN (Max Sequence Length: 64)

Total misclassified examples: 20

Example 1 *(Index 3: Gold: Sci/Tech, Predicted: Sports)*

```
Prediction Unit Helps Forecast Wildfires (AP) AP - It's barely dawn when Mike Fitzpatrick starts his shift with a blur of colorful maps, figures and endless
↪ charts, but already he knows what the day will bring. Lightning will strike in places he expects. Winds will pick up, moist places will dry and flames will
↪ roar.
```

Example 2 *(Index 8: Gold: Sci/Tech, Predicted: World)*

```
E-mail scam targets police chief Wiltshire Police warns about "phishing" after its fraud squad chief was targeted.
```

Example 3 *(Index 9: Gold: Sci/Tech, Predicted: Business)*

```
Card fraud unit nets 36,000 cards In its first two years, the UK's dedicated card fraud unit, has recovered 36,000 stolen cards and 171 arrests - and estimates
↪ it saved 65m.
```

Example 4 *(Index 13: Gold: Sci/Tech, Predicted: World)*

```
Super ant colony hits Australia A giant 100km colony of ants which has been discovered in Melbourne, Australia, could threaten local insect species.
```

Example 5 *(Index 18: Gold: Sci/Tech, Predicted: Business)*

```
Rocking the Cradle of Life When did life begin? One evidential clue stems from the fossil records in Western Australia, although whether these layered sediments
↪ are biological or chemical has spawned a spirited debate. Oxford researcher, Nicola McLoughlin, describes some of the issues in contention.
```

Example 6 *(Index 20: Gold: Sci/Tech, Predicted: Business)*

```
IBM to hire even more new workers By the end of the year, the computing giant plans to have its biggest headcount since 1991.
```

Example 7 *(Index 23: Gold: Sci/Tech, Predicted: Business)*

```
Some People Not Eligible to Get in on Google IPO Google has billed its IPO as a way for everyday people to get in on the process, denying Wall Street the usual
↪ stranglehold it's had on IPOs. Public bidding, a minimum of just five shares, an open process with 28 underwriters - all this pointed to a new level of
↪ public participation. But this isn't the case.
```

Example 8 *(Index 36: Gold: World, Predicted: Business)*

```
Venezuela Prepares for Chavez Recall Vote Supporters and rivals warn of possible fraud; government says Chavez's defeat could produce turmoil in world oil
↪ market.
```

Example 9 *(Index 79: Gold: World, Predicted: Sports)*

```
Live: Olympics day four Richard Faulds and Stephen Parry are going for gold for Great Britain on day four in Athens.
```

Example 10 *(Index 83: Gold: Business, Predicted: Sci/Tech)*

```
Intel to delay product aimed for high-definition TVs SAN FRANCISCO -- In the latest of a series of product delays, Intel Corp. has postponed the launch of a
↪ video display chip it had previously planned to introduce by year end, putting off a showdown with Texas Instruments Inc. in the fast-growing market for
↪ high-definition television displays.
```

Example 11 *(Index 110: Gold: Business, Predicted: Sci/Tech)*

```
Yahoo! Ups Ante for Small Businesses Web giant Yahoo! is gambling that price cuts on its domain name registration and Web hosting products will make it more
↪ competitive with discounters in the space -- which means that small businesses looking to move online get a sweeter deal through ...
```

Example 12 *(Index 120: Gold: World, Predicted: Business)*

```
Oil prices bubble to record high The price of oil has continued its sharp rise overnight, closing at a record high. The main contract in New York, light sweet
↪ crude for delivery next month, has closed at a record $US46.75 a barrel - up 70 cents on yesterday 's close.
```

Example 13 *(Index 155: Gold: World, Predicted: Sci/Tech)*

```
Future Doctors, Crossing Borders Students at the Mount Sinai School of Medicine learn that diet and culture shape health in East Harlem.
```

Example 14 *(Index 168: Gold: Sports, Predicted: World)*

Afghan women make brief Olympic debut Afghan women made a short-lived debut in the Olympic Games on Wednesday as 18-year-old judo wildcard Friba Razayee was defeated after 45 seconds of her first match in the under-70kg middleweight.

Example 15 *(Index 178: Gold: Sci/Tech, Predicted: Business)*

Oracle launches Business Integlligence 10g Oracle introduced a new BI platform yesterday, Business Intelligence 10g that rolls up into one solution all of their BI tools. However, more interesting than the nitty-gritty details of what is included is the back story taking place at the same time.

Example 16 *(Index 196: Gold: World, Predicted: Business)*

Stock Prices Climb Ahead of Google IPO NEW YORK - Investors shrugged off rising crude futures Wednesday to capture well-priced shares, sending the Nasdaq composite index up 1.6 percent ahead of Google Inc.'s much-anticipated initial public offering of stock. In afternoon trading, the Dow Jones industrial average gained 67.10, or 0.7 percent, to 10,039.93...

Example 17 *(Index 207: Gold: Sci/Tech, Predicted: Business)*

Intuit Posts Wider Loss After Charge (Reuters) Reuters - Intuit Inc. (INTU.O), maker of the No. 1 U.S. tax presentation software TurboTax, on Wednesday posted a wider quarterly loss after taking a goodwill impairment charge during its seasonally weaker fourth quarter.

Example 18 *(Index 215: Gold: Sci/Tech, Predicted: Business)*

Hurricane center 's projection on Charley not far off, data show FORT LAUDERDALE, Fla. - (KRT) - Despite criticism that it should have better anticipated Hurricane Charley 's rapid intensification and quick turn, the National Hurricane Center 's forecast wasn 't that far off, a preliminary post-mortem shows.

Example 19 *(Index 222: Gold: Business, Predicted: World)*

Selling Houston Warts and All, Especially Warts Descriptions of urban afflictions and images of giant mosquitoes and cockroaches to convey a sense of how Houston is nevertheless beloved by many residents.

Example 20 *(Index 240: Gold: Sci/Tech, Predicted: Business)*

Bill Clinton Helps Launch Search Engine Former president Bill Clinton on Monday helped launch a new Internet search company backed by the Chinese government which says its technology uses artificial intelligence to produce better results than Google Inc.

A.2 CNN (Max Sequence Length: 128)

Total misclassified examples: 20

Example 1 *(Index 3: Gold: Sci/Tech, Predicted: Sports)*

```
Prediction Unit Helps Forecast Wildfires (AP) AP - It's barely dawn when Mike Fitzpatrick starts his shift with a blur of colorful maps, figures and endless
↪ charts, but already he knows what the day will bring. Lightning will strike in places he expects. Winds will pick up, moist places will dry and flames will
↪ roar.
```

Example 2 *(Index 8: Gold: Sci/Tech, Predicted: World)*

```
E-mail scam targets police chief Wiltshire Police warns about "phishing" after its fraud squad chief was targeted.
```

Example 3 *(Index 13: Gold: Sci/Tech, Predicted: World)*

```
Super ant colony hits Australia A giant 100km colony of ants which has been discovered in Melbourne, Australia, could threaten local insect species.
```

Example 4 *(Index 23: Gold: Sci/Tech, Predicted: Business)*

```
Some People Not Eligible to Get in on Google IPO Google has billed its IPO as a way for everyday people to get in on the process, denying Wall Street the usual
↪ stranglehold it's had on IPOs. Public bidding, a minimum of just five shares, an open process with 28 underwriters - all this pointed to a new level of
↪ public participation. But this isn't the case.
```

Example 5 *(Index 28: Gold: Sports, Predicted: World)*

```
Shoppach doesn't appear ready to hit the next level With the weeks dwindling until Jason Varitek enters free agency, the Red Sox continue to carefully monitor
↪ Kelly Shoppach , their catcher of the future, in his climb toward the majors. The Sox like most of what they have seen at Triple A Pawtucket from Shoppach,
↪ though it remains highly uncertain whether he can make the adjustments at the plate ...
```

Example 6 *(Index 36: Gold: World, Predicted: Business)*

```
Venezuela Prepares for Chavez Recall Vote Supporters and rivals warn of possible fraud; government says Chavez's defeat could produce turmoil in world oil
↪ market.
```

Example 7 *(Index 41: Gold: Business, Predicted: Sci/Tech)*

Retailers Vie for Back-To-School Buyers (Reuters) Reuters - Apparel retailers are hoping their back-to-school fashions will make the grade among style-conscious teens and young adults this fall, but it could be a tough sell, with students and parents keeping a tighter hold on their wallets.

Example 8 *(Index 79: Gold: World, Predicted: Sports)*

Live: Olympics day four Richard Faulds and Stephen Parry are going for gold for Great Britain on day four in Athens.

Example 9 *(Index 83: Gold: Business, Predicted: Sci/Tech)*

Intel to delay product aimed for high-definition TVs SAN FRANCISCO -- In the latest of a series of product delays, Intel Corp. has postponed the launch of a video display chip it had previously planned to introduce by year end, putting off a showdown with Texas Instruments Inc. in the fast-growing market for high-definition television displays.

Example 10 *(Index 110: Gold: Business, Predicted: Sci/Tech)*

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Oil prices bubble to record high The price of oil has continued its sharp rise overnight, closing at a record high. The main contract in New York, light sweet crude for delivery next month, has closed at a record $US46.75 a barrel - up 70 cents on yesterday 's close.

Example 12 *(Index 155: Gold: World, Predicted: Business)*

Future Doctors, Crossing Borders Students at the Mount Sinai School of Medicine learn that diet and culture shape health in East Harlem.

Example 13 *(Index 168: Gold: Sports, Predicted: World)*

Afghan women make brief Olympic debut Afghan women made a short-lived debut in the Olympic Games on Wednesday as 18-year-old judo wildcard Friba Razayee was defeated after 45 seconds of her first match in the under-70kg middleweight.

Example 14 *(Index 178: Gold: Sci/Tech, Predicted: Business)*

Oracle launches Business Integlligence 10g Oracle introduced a new BI platform yesterday, Business Intelligence 10g that rolls up into one solution all of their BI tools. However, more interesting than the nitty-gritty details of what is included is the back story taking place at the same time.

Example 15 *(Index 183: Gold: Sci/Tech, Predicted: Business)*

```
Microsoft finalises three-year government deal Hot on the heels of its 10-year strategic partnership with the London Borough of Newham, Microsoft is close to
↪ signing a new broad three-year public sector agreement with the government.
```

Example 16 *(Index 207: Gold: Sci/Tech, Predicted: Business)*

```
Intuit Posts Wider Loss After Charge (Reuters) Reuters - Intuit Inc. (INTU.O), maker of the No. 1 U.S. tax presentation software TurboTax, on Wednesday posted a
↪ wider quarterly loss after taking a goodwill impairment charge during its seasonally weaker fourth quarter.
```

Example 17 *(Index 208: Gold: Sports, Predicted: World)*

```
Hamilton Wins Cycling Time Trial Event THENS, Aug. 18 Tyler Hamilton had bruises splotched all over his back, painful souvenirs of a Tour de France gone terribly
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```

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```

Example 19 *(Index 222: Gold: Business, Predicted: World)*

```
Selling Houston Warts and All, Especially Warts Descriptions of urban afflictions and images of giant mosquitoes and cockroaches to convey a sense of how Houston
↪ is nevertheless beloved by many residents.
```

Example 20 *(Index 240: Gold: Sci/Tech, Predicted: Business)*

```
Bill Clinton Helps Launch Search Engine Former president Bill Clinton on Monday helped launch a new Internet search company backed by the Chinese government
↪ which says its technology uses artificial intelligence to produce better results than Google Inc.
```

A.3 CNN (Max Sequence Length: 256)

Total misclassified examples: 20

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```
Prediction Unit Helps Forecast Wildfires (AP) AP - It's barely dawn when Mike Fitzpatrick starts his shift with a blur of colorful maps, figures and endless
↪ charts, but already he knows what the day will bring. Lightning will strike in places he expects. Winds will pick up, moist places will dry and flames will
↪ roar.
```

Example 2 *(Index 8: Gold: Sci/Tech, Predicted: World)*

```
E-mail scam targets police chief Wiltshire Police warns about "phishing" after its fraud squad chief was targeted.
```

Example 3 *(Index 9: Gold: Sci/Tech, Predicted: Business)*

```
Card fraud unit nets 36,000 cards In its first two years, the UK's dedicated card fraud unit, has recovered 36,000 stolen cards and 171 arrests - and estimates
↪ it saved 65m.
```

Example 4 *(Index 13: Gold: Sci/Tech, Predicted: World)*

```
Super ant colony hits Australia A giant 100km colony of ants which has been discovered in Melbourne, Australia, could threaten local insect species.
```

Example 5 *(Index 20: Gold: Sci/Tech, Predicted: Business)*

```
IBM to hire even more new workers By the end of the year, the computing giant plans to have its biggest headcount since 1991.
```

Example 6 *(Index 36: Gold: World, Predicted: Business)*

```
Venezuela Prepares for Chavez Recall Vote Supporters and rivals warn of possible fraud; government says Chavez's defeat could produce turmoil in world oil
↪ market.
```

Example 7 *(Index 55: Gold: Sci/Tech, Predicted: Sports)*

```
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```

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```
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```

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Example 12 *(Index 139: Gold: World, Predicted: Business)*

Sharon orders 1,000 homes in West Bank Israel announced plans for 1,000 houses in the West Bank yesterday, accelerating the expansion of the settlements.

Example 13 *(Index 155: Gold: World, Predicted: Business)*

Future Doctors, Crossing Borders Students at the Mount Sinai School of Medicine learn that diet and culture shape health in East Harlem.

Example 14 *(Index 207: Gold: Sci/Tech, Predicted: Business)*

Intuit Posts Wider Loss After Charge (Reuters) Reuters - Intuit Inc. (INTU.O), maker of the No. 1 U.S. tax presentation software TurboTax, on Wednesday posted a wider quarterly loss after taking a goodwill impairment charge during its seasonally weaker fourth quarter.

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Example 17 *(Index 240: Gold: Sci/Tech, Predicted: Business)*

```
Bill Clinton Helps Launch Search Engine Former president Bill Clinton on Monday helped launch a new Internet search company backed by the Chinese government
↪ which says its technology uses artificial intelligence to produce better results than Google Inc.
```

Example 18 *(Index 258: Gold: Sci/Tech, Predicted: World)*

```
Ciena Posts a Loss, Forecasts Flat Sales <p></p><p> By Deborah Cohen</p><p> CHICAGO (Reuters) - Telecommunications equipment makerCiena Corp. <CIEN.O> on
↪ Thursday reported a wider loss for thefiscal third quarter due to slack demand and forecast sales inthe current quarter would be little changed from the
↪ thirdquarter.</p>
```

Example 19 *(Index 262: Gold: Sci/Tech, Predicted: Business)*

```
Ecuadorean Lawsuit Vs Texaco Boils Down to Science (Reuters) Reuters - After a decade of court battles, lawyers on Wednesday took a lawsuit by Ecuadorean Indians
↪ accusing U.S. oil firm ChevronTexaco Corp.. of polluting the Amazon jungle into the field.
```

Example 20 *(Index 267: Gold: Sci/Tech, Predicted: Business)*

```
Court: File-swapping software not liable for copyright violations The makers of two leading file-sharing programs are not legally liable for the songs, movies
↪ and other copyright works swapped online by their users, a federal appeals court ruled Thursday in a stinging blow to the entertainment industry.
```

A.4 LSTM (Max Sequence Length: 64)

Total misclassified examples: 20

Example 1 *(Index 3: Gold: Sci/Tech, Predicted: Sports)*

```
Prediction Unit Helps Forecast Wildfires (AP) AP - It's barely dawn when Mike Fitzpatrick starts his shift with a blur of colorful maps, figures and endless
↪ charts, but already he knows what the day will bring. Lightning will strike in places he expects. Winds will pick up, moist places will dry and flames will
↪ roar.
```

Example 2 *(Index 7: Gold: Sci/Tech, Predicted: Sports)*

```
FOAFKey: FOAF, PGP, Key Distribution, and Bloom Filters FOAF/LOAF and bloom filters have a lot of interesting properties for social network and whitelist
↪ distribution. I think we can go one level higher though and include GPG/OpenPGP key fingerpring distribution in the FOAF file for simple web-of-trust based
↪ key distribution. What if we used FOAF and included the PGP key fingerprint(s) for identities? This could mean a lot. You include the PGP key fingerprints
↪ within the FOAF file of your direct friends and then include a bloom filter of the PGP key fingerprints of your entire whitelist (the source FOAF file would
↪ of course need to be encrypted ). Your whitelist would be populated from the social network as your client discovered new identit ...
```

Example 3 *(Index 9: Gold: Sci/Tech, Predicted: Business)*

```
Card fraud unit nets 36,000 cards In its first two years, the UK's dedicated card fraud unit, has recovered 36,000 stolen cards and 171 arrests - and estimates
↪ it saved 65m.
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Example 4 *(Index 20: Gold: Sci/Tech, Predicted: Business)*

```
IBM to hire even more new workers By the end of the year, the computing giant plans to have its biggest headcount since 1991.
```

Example 5 *(Index 23: Gold: Sci/Tech, Predicted: Business)*

```
Some People Not Eligible to Get in on Google IPO Google has billed its IPO as a way for everyday people to get in on the process, denying Wall Street the usual
↪ stranglehold it's had on IPOs. Public bidding, a minimum of just five shares, an open process with 28 underwriters - all this pointed to a new level of
↪ public participation. But this isn't the case.
```

Example 6 *(Index 24: Gold: Sci/Tech, Predicted: Business)*

```
Rivals Try to Turn Tables on Charles Schwab By MICHAEL LIEDTKE SAN FRANCISCO (AP) -- With its low prices and iconoclastic attitude, discount stock broker Charles
↪ Schwab Corp. (SCH) represented an annoying stone in Wall Street's wing-tipped shoes for decades...
```

Example 7 *(Index 33: Gold: World, Predicted: Sci/Tech)*

```
Man Sought $50M From McGreevey, Aides Say (AP) AP - The man who claims Gov. James E. McGreevey sexually harassed him was pushing for a cash settlement of up to
↪ $50 million before the governor decided to announce that he was gay and had an extramarital affair, sources told The Associated Press.
```

Example 8 *(Index 55: Gold: Sci/Tech, Predicted: World)*

```
Promoting a Shared Vision As Michael Kaleko kept running into people who were getting older and having more vision problems, he realized he could do something
↪ about it.
```

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```
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↪ video display chip it had previously planned to introduce by year end, putting off a showdown with Texas Instruments Inc. in the fast-growing market for
↪ high-definition television displays.
```

Example 10 *(Index 102: Gold: Sci/Tech, Predicted: World)*

China cracks down on quot;phone sex quot; services BEIJING, Aug. 17 (Xinhuanet) -- China is carrying out a nationwide campaign to crack down on quot;phone sex quot; services, paralleling another sweeping operation against Internet pornography, Minister of Information Industry Wang Xudong said here Tuesday.

Example 11 *(Index 110: Gold: Business, Predicted: Sci/Tech)*

Yahoo! Ups Ante for Small Businesses Web giant Yahoo! is gambling that price cuts on its domain name registration and Web hosting products will make it more competitive with discounters in the space -- which means that small businesses looking to move online get a sweeter deal through ...

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Oil prices bubble to record high The price of oil has continued its sharp rise overnight, closing at a record high. The main contract in New York, light sweet crude for delivery next month, has closed at a record $US46.75 a barrel - up 70 cents on yesterday 's close.

Example 13 *(Index 121: Gold: Sports, Predicted: World)*

Notable quotes Tuesday at the Athens Olympics quot;It hurt like hell. I could see (Thorpe) coming up. But when I was breathing, I saw my team going crazy -- and that really kept me going. quot; ...

Example 14 *(Index 125: Gold: Business, Predicted: World)*

SUVs: Live And Let Die NEW YORK - The newly released traffic crash fatality data have something for everyone in the debate about the safety of sport utility vehicles.

Example 15 *(Index 126: Gold: Sports, Predicted: World)*

Security scare as intruder dives in A CANADIAN husband 's love for his wife has led to a tightening of security at all Olympic venues in Athens.

Example 16 *(Index 155: Gold: World, Predicted: Business)*

Future Doctors, Crossing Borders Students at the Mount Sinai School of Medicine learn that diet and culture shape health in East Harlem.

Example 17 *(Index 188: Gold: Sports, Predicted: Sci/Tech)*

Veterans Committee counts for little The Hall of Fame released the latest Veterans Committee ballot yesterday. As you might (or might not) remember, there 's a (nearly) new committee in town.

Example 18 *(Index 206: Gold: Sports, Predicted: World)*

```
Boro Captain Warns of Duo 's Threat Gareth Southgate has warned Barclays Premiership defences to be wary of Middlesbroughs back-to-form strikers Mark Viduka and
↪ Jimmy Floyd Hasselbaink.
```

Example 19 *(Index 208: Gold: Sports, Predicted: World)*

```
Hamilton Wins Cycling Time Trial Event THENS, Aug. 18 Tyler Hamilton had bruises splotched all over his back, painful souvenirs of a Tour de France gone terribly
↪ wrong.
```

Example 20 *(Index 215: Gold: Sci/Tech, Predicted: Business)*

```
Hurricane center 's projection on Charley not far off, data show FORT LAUDERDALE, Fla. - (KRT) - Despite criticism that it should have better anticipated
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```

A.5 LSTM (Max Sequence Length: 128)

Total misclassified examples: 20

Example 1 *(Index 20: Gold: Sci/Tech, Predicted: Business)*

```
IBM to hire even more new workers By the end of the year, the computing giant plans to have its biggest headcount since 1991.
```

Example 2 *(Index 23: Gold: Sci/Tech, Predicted: Business)*

```
Some People Not Eligible to Get in on Google IPO Google has billed its IPO as a way for everyday people to get in on the process, denying Wall Street the usual
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```

Example 4 *(Index 55: Gold: Sci/Tech, Predicted: World)*

Promoting a Shared Vision As Michael Kaleko kept running into people who were getting older and having more vision problems, he realized he could do something about it.

Example 5 *(Index 77: Gold: Sports, Predicted: Business)*

Griffin to Anchor D-Line The Redskins expect huge things from 300-pound Cornelius Griffin, who was signed to aid the team's weakest unit - the defensive line.

Example 6 *(Index 83: Gold: Business, Predicted: Sci/Tech)*

Intel to delay product aimed for high-definition TVs SAN FRANCISCO -- In the latest of a series of product delays, Intel Corp. has postponed the launch of a video display chip it had previously planned to introduce by year end, putting off a showdown with Texas Instruments Inc. in the fast-growing market for high-definition television displays.

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Example 9 *(Index 125: Gold: Business, Predicted: Sci/Tech)*

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Example 10 *(Index 126: Gold: Sports, Predicted: World)*

Security scare as intruder dives in A CANADIAN husband 's love for his wife has led to a tightening of security at all Olympic venues in Athens.

Example 11 *(Index 133: Gold: World, Predicted: Sci/Tech)*

Kerry-Kerrey Confusion Trips Up Campaign (AP) AP - John Kerry, Bob Kerrey. It's easy to get confused.

Example 12 *(Index 155: Gold: World, Predicted: Business)*

```
Future Doctors, Crossing Borders Students at the Mount Sinai School of Medicine learn that diet and culture shape health in East Harlem.
```

Example 13 *(Index 173: Gold: Sci/Tech, Predicted: World)*

```
Small computers can have multiple personalities, too BOSTON The jury is still out on whether a computer can ever truly be intelligent, but there is no question
↪ that it can have multiple personalities. It 's just a matter of software. We usually think of the processor chip as the brains of a computer. The ...
```

Example 14 *(Index 188: Gold: Sports, Predicted: Sci/Tech)*

```
Veterans Committee counts for little The Hall of Fame released the latest Veterans Committee ballot yesterday. As you might (or might not) remember, there 's a
↪ (nearly) new committee in town.
```

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```
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```

Example 17 *(Index 218: Gold: Sci/Tech, Predicted: Business)*

```
Lloyds TSB cashes in on VoIP Lloyds TSB is gearing up to roll out one of the largest converged networks in Europe, a 500m 70,000 phone VoIP infrastructure
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```

Example 18 *(Index 222: Gold: Business, Predicted: Sci/Tech)*

```
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↪ is nevertheless beloved by many residents.
```

Example 19 *(Index 240: Gold: Sci/Tech, Predicted: Business)*

```
Bill Clinton Helps Launch Search Engine Former president Bill Clinton on Monday helped launch a new Internet search company backed by the Chinese government
↪ which says its technology uses artificial intelligence to produce better results than Google Inc.
```

Example 20 *(Index 241: Gold: Sports, Predicted: World)*

```
Harris ' three-run double in ninth sinks Gagne LOS ANGELES - Paul Lo Duca never got to moonwalk to home plate, though he did skip gleefully to the dugout moments
↪ after facing former batterymate Eric Gagne for the first time.
```

A.6 LSTM (Max Sequence Length: 256)

Total misclassified examples: 20

Example 1 *(Index 3: Gold: Sci/Tech, Predicted: Sports)*

```
Prediction Unit Helps Forecast Wildfires (AP) AP - It's barely dawn when Mike Fitzpatrick starts his shift with a blur of colorful maps, figures and endless
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E-mail scam targets police chief Wiltshire Police warns about "phishing" after its fraud squad chief was targeted.
```

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```
Security scare as intruder dives in A CANADIAN husband 's love for his wife has led to a tightening of security at all Olympic venues in Athens.
```

Example 11 *(Index 154: Gold: World, Predicted: Business)*

```
Google Lowers Its IPO Price Range SAN JOSE, Calif. - In a sign that Google Inc.'s initial public offering isn't as popular as expected, the company lowered its
↪ estimated price range to between $85 and $95 per share, down from the earlier prediction of $108 and $135 per share...
```

Example 12 *(Index 155: Gold: World, Predicted: Business)*

```
Future Doctors, Crossing Borders Students at the Mount Sinai School of Medicine learn that diet and culture shape health in East Harlem.
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Example 15 *(Index 188: Gold: Sports, Predicted: World)*

Veterans Committee counts for little The Hall of Fame released the latest Veterans Committee ballot yesterday. As you might (or might not) remember, there 's a (nearly) new committee in town.

Example 16 *(Index 196: Gold: World, Predicted: Business)*

Stock Prices Climb Ahead of Google IPO NEW YORK - Investors shrugged off rising crude futures Wednesday to capture well-priced shares, sending the Nasdaq composite index up 1.6 percent ahead of Google Inc.'s much-anticipated initial public offering of stock. In afternoon trading, the Dow Jones industrial average gained 67.10, or 0.7 percent, to 10,039.93...

Example 17 *(Index 207: Gold: Sci/Tech, Predicted: Business)*

Intuit Posts Wider Loss After Charge (Reuters) Reuters - Intuit Inc. (INTU.O), maker of the No. 1 U.S. tax presentation software TurboTax, on Wednesday posted a wider quarterly loss after taking a goodwill impairment charge during its seasonally weaker fourth quarter.

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Selling Houston Warts and All, Especially Warts Descriptions of urban afflictions and images of giant mosquitoes and cockroaches to convey a sense of how Houston is nevertheless beloved by many residents.

B Classification Reports and Confusion Matrices

This appendix provides the full test-set classification report and confusion matrix for each experimental run.

B.1 CNN (Max Sequence Length: 64)

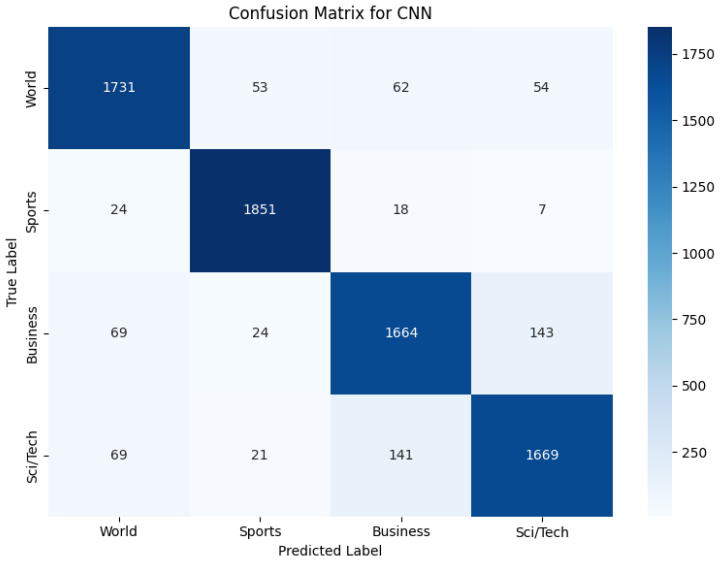

Figure 5: Confusion matrix for CNN with maximum sequence length 64.

```
              precision    recall  f1-score   support

           0     0.9144    0.9111    0.9127      1900
           1     0.9497    0.9742    0.9618      1900
           2     0.8828    0.8758    0.8793      1900
           3     0.8911    0.8784    0.8847      1900

    accuracy                         0.9099      7600
   macro avg     0.9095    0.9099    0.9096      7600
weighted avg     0.9095    0.9099    0.9096      7600
```

B.2 CNN (Max Sequence Length: 128)

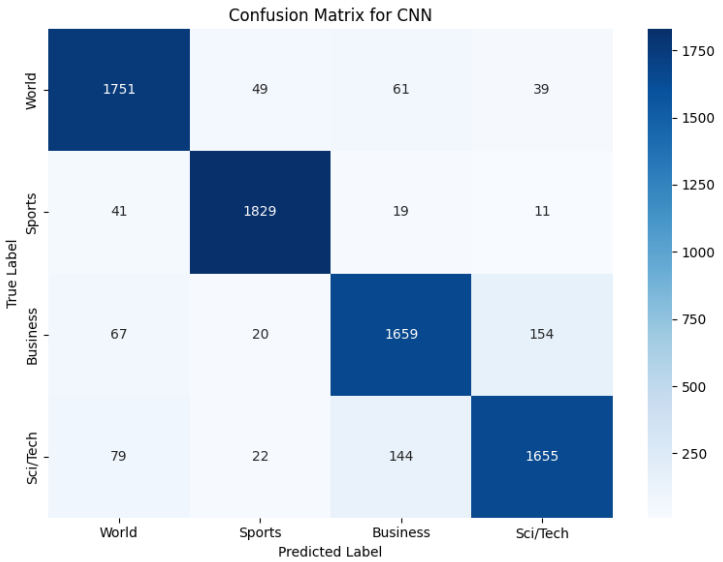

Figure 6: Confusion matrix for CNN with maximum sequence length 128.

```
              precision    recall  f1-score   support

           0     0.9035    0.9216    0.9125      1900
           1     0.9526    0.9626    0.9576      1900
           2     0.8810    0.8732    0.8771      1900
           3     0.8903    0.8711    0.8806      1900

    accuracy                         0.9071      7600
   macro avg     0.9069    0.9071    0.9069      7600
weighted avg     0.9069    0.9071    0.9069      7600
```

B.3 CNN (Max Sequence Length: 256)

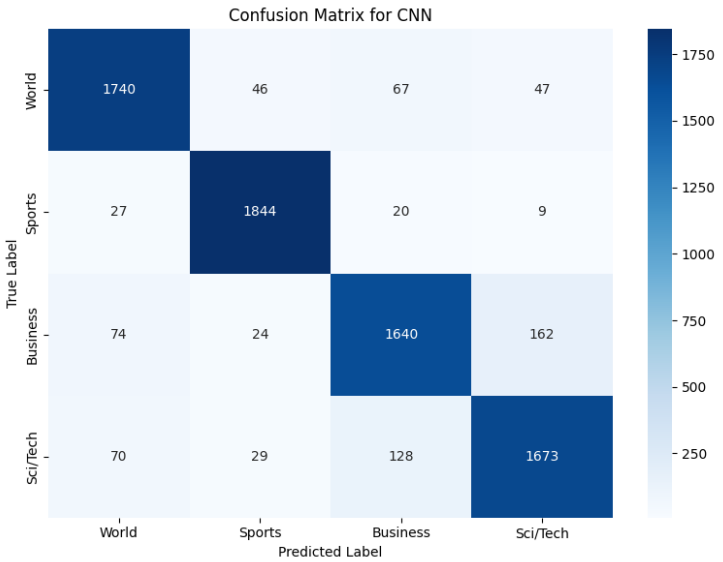

Figure 7: Confusion matrix for CNN with maximum sequence length 256.

```
              precision    recall  f1-score   support

           0     0.9105    0.9158    0.9131      1900
           1     0.9490    0.9705    0.9597      1900
           2     0.8841    0.8632    0.8735      1900
           3     0.8847    0.8805    0.8826      1900

    accuracy                         0.9075      7600
   macro avg     0.9071    0.9075    0.9072      7600
weighted avg     0.9071    0.9075    0.9072      7600
```

B.4 LSTM (Max Sequence Length: 64)

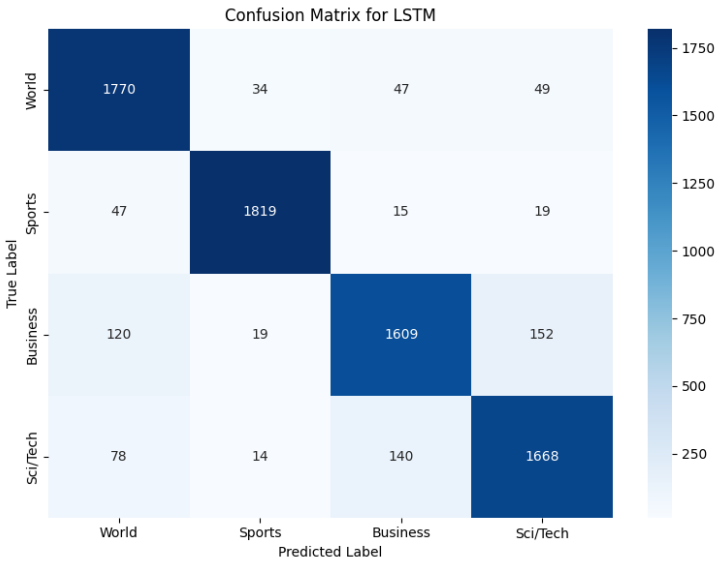

Figure 8: Confusion matrix for LSTM with maximum sequence length 64.

	precision	recall	f1-score	support
0	0.8784	0.9316	0.9042	1900
1	0.9645	0.9574	0.9609	1900
2	0.8885	0.8468	0.8672	1900
3	0.8835	0.8779	0.8807	1900
accuracy			0.9034	7600
macro avg	0.9037	0.9034	0.9032	7600
weighted avg	0.9037	0.9034	0.9032	7600

B.5 LSTM (Max Sequence Length: 128)

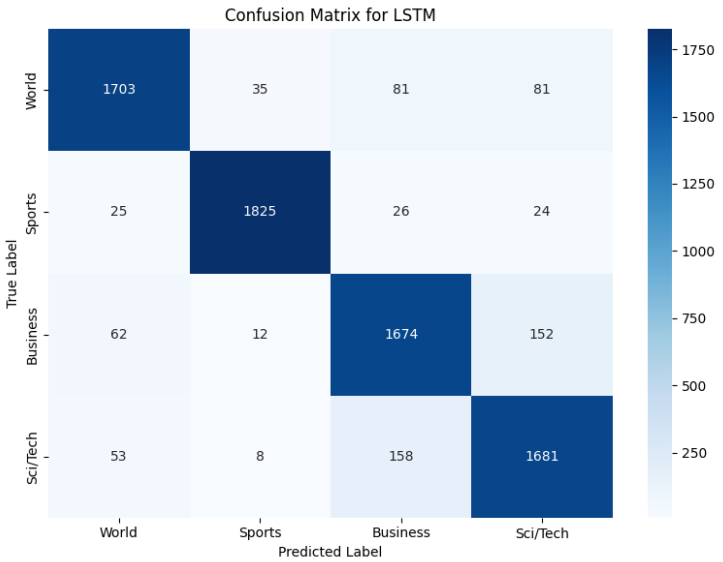

Figure 9: Confusion matrix for LSTM with maximum sequence length 128.

	precision	recall	f1-score	support
0	0.9240	0.8963	0.9100	1900
1	0.9707	0.9605	0.9656	1900
2	0.8633	0.8811	0.8721	1900
3	0.8674	0.8847	0.8760	1900
accuracy			0.9057	7600
macro avg	0.9064	0.9057	0.9059	7600
weighted avg	0.9064	0.9057	0.9059	7600

B.6 LSTM (Max Sequence Length: 256)

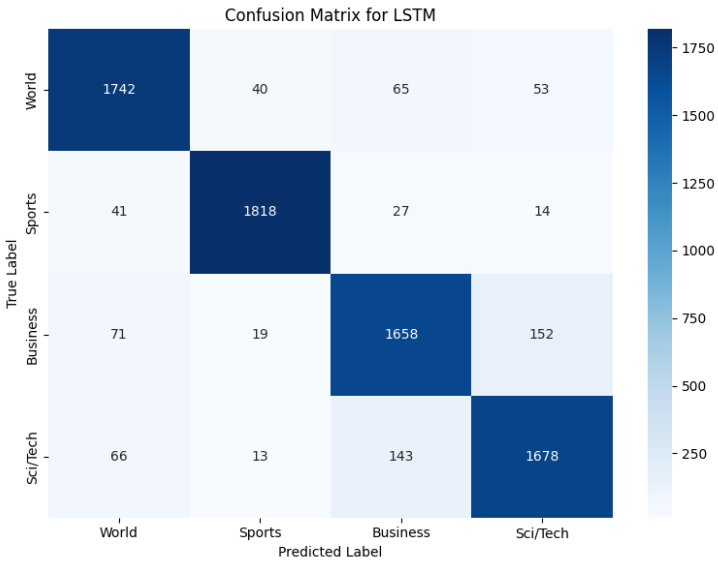

Figure 10: Confusion matrix for LSTM with maximum sequence length 256.

```
              precision    recall  f1-score   support

           0     0.9073    0.9168    0.9120      1900
           1     0.9619    0.9568    0.9594      1900
           2     0.8759    0.8726    0.8742      1900
           3     0.8846    0.8832    0.8839      1900

    accuracy                         0.9074      7600
   macro avg     0.9074    0.9074    0.9074      7600
weighted avg     0.9074    0.9074    0.9074      7600
```

C Additional Figures

This appendix contains additional figures that were generated during the analysis but were not included in the main report due to space constraints.

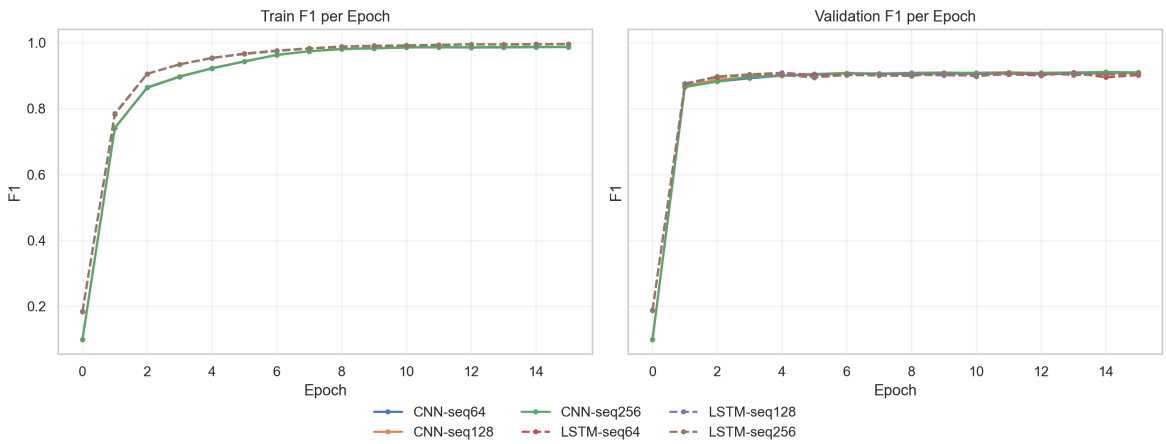

Figure 11: F1 score curves for training and validation sets across epochs for all six runs.